

VAJRALA GEETHIKA

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Summary

Data-oriented professional with a strong foundation in Python, classical machine learning, and Generative AI. Experienced in developing and evaluating machine learning models through rigorous data cleaning and feature engineering. Skilled in implementing LLM prompt engineering techniques, including few-shot and chain-of-thought methods, to enhance output quality. Committed to supporting engineering teams through systematic model testing, benchmarking, and collaborative documentation practices.

Technical Skills

Programming:	Python (Scikit-Learn, PyTorch, Pandas), SQL, R
Machine Learning:	Supervised Learning, Regression, Classification, Model Evaluation, Benchmarking
AI & LLM:	Prompt Engineering, Few-shot Prompting, Chain-of-Thought, Generative AI
Data & Tools:	Git, VS Code, Jupyter Notebook, MySQL, PostgreSQL
Visualization:	Power BI, Tableau, Advanced Excel

PROJECTS

Loan Default Risk Analysis Pipeline

Python, Pandas, SQL, ML

- Developed classification models using Scikit-Learn to predict default risk, improving predictive accuracy by 15% through optimized feature selection.
- Implemented data preprocessing pipelines for 50,000+ customer records to ensure consistent training data quality.
- Collaborated with engineering teams to conduct offline evaluations and benchmarking of model outputs against baseline metrics.
- Wrote scripts for automated validation of model performance metrics to facilitate weekly sprint planning reviews.
- Assisted in documenting experimental results and model hyperparameters to maintain clear version history.

Customer Purchase Behavior Analytics

SQL, Power BI, Excel

- Developed SQL-based data extraction workflows for 3,000+ transaction records, supporting large-scale dataset preparation for machine learning.
- Implemented automated data cleaning routines in Python, reducing manual labor requirements for feature engineering by 10 hours per week.
- Collaborated with cross-functional teams to define data requirements and benchmark business KPI tracking metrics.
- Created visualizations to interpret model predictions and communicate findings to stakeholders during model reviews.
- Supported the deployment of data pipelines by testing integration points and verifying data integrity at each stage.

AI-Powered Computer Vision Attendance System

Python, MySQL, OpenCV

- Developed a computer vision pipeline for facial recognition, successfully increasing identification accuracy by 25% through iterative tuning.
- Implemented pre-processing steps for image datasets, ensuring reliable model performance across varied environmental lighting conditions.
- Optimized database integration to support 1,000+ records with sub-second retrieval latency for real-time inference.
- Tested model robustness by simulating edge cases and documenting failure modes for future system improvements.
- Contributed to project documentation and weekly progress briefings by summarizing technical roadblocks and milestones.

EDUCATION

SASTRA Deemed University | Bachelor of Technology in Electronics and Communication Engineering

Thanjavur, Tamil Nadu

Achievements: CGPA: 7.5 / 10

CERTIFICATIONS

Microsoft Certified: Python for Data Science

Scaler Certified: SQL & Database Design with MySQL